

Target Incident: TAYLOR_ENERGY Incident

Satellite Sensor: Sentinel-1 SAR (10m C-band)

Report Date: 2026-09-28 UTC

PRIMARY VERDICT SUSPECTED_ANCHOR_STRIKE_ON_PIPELINE	ATTRIBUTION CONFIDENCE HIGH (88.5%)	GROUND-TRUTH OFFSET 309 m	VESSELS EXONERATED 15 / 15 (100%)
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Official Statutory Forensic Finding

Executive Finding: Vessel CG WALNUT (MMSI: 366953000) was observed loitering/anchoring within 119m of Pipeline Segment 12712 operated by WALTER OIL & GAS CORPORATION prior to release.

Reconstructed Origin: 28.93537°N, -88.96780°W | Spill Release Window: 2023-11-17T23:16:46+00:00 to 2023-11-17T23:24:16+00:00

Scientific Benchmark Ground-Truth Verification

Known Facility: Taylor Energy MC-20 Wellhead Site (BSEE / NOAA Official)

Official Ground Truth: 28.93653°N, -88.97069°W

Model Predicted Origin: 28.93537°N, -88.96780°W

Absolute Offset: 309.1 meters (0.167 nm)

Validation Status: ACCURATE (Sub-quarter-mile precision)

In September 2004, Hurricane Ivan triggered an underwater mudslide that toppled the Taylor Energy Mississippi Canyon Block 20 production platform, pulling the jacket into 475 feet of water and burying 28 wellheads beneath 100 feet of sediment. The site produces a continuous, chronic hydrocarbon discharge. Since April 2019, an offshore acoustic subsea containment dome system operated by the Couvillion Group under USCG / BSEE oversight captures and recovers approximately 1,000 gallons per day of crude oil, though persistent residual sheen continuously emanates from surrounding seabed seeps.

1. SATELLITE REMOTE SENSING & SLICK MORPHOLOGY ANALYSIS

Morphological Metric	Calculated Value	Forensic & Physical Significance
Satellite Overpass Time	2023-11-17T23:54:16+00:00	Copernicus Sentinel-1 Synthetic Aperture Radar (SAR) acquisition
Detected Slick Centroid	28.93471°N, -88.96548°W	Center of gravity of low-backscatter oil film polygon
Major Spreading Axis Length	2.67 km	Longitudinal extent along principal dispersion direction
Transverse Width Dimensions	Head: 1780 m Tail: 2136 m	Diffusion profile: $W_{tail} > W_{head}$ confirms continuous drift away from source
Calculated Drift Age (ΔT)	0.50 hours (30 min)	Fickian turbulent diffusion model (horizontal diffusivity $K_h = 10 \text{ m}^2/\text{s}$)

Fickian Diffusion Mathematical Formulation:

Under 2D Gaussian turbulent diffusion, plume width expands as $W(t)^2 = 8K_h t$. Solving for elapsed drift time ΔT from transverse asymmetry yields:

$$\Delta T = (W_{tail}^2 - W_{head}^2) \div (8 \times K_h)$$

Substituting $W_{tail} = 2136.1 \text{ m}$, $W_{head} = 1780.1 \text{ m}$, and $K_h = 10.0 \text{ m}^2/\text{s}$ yields $\Delta T = 0.50 \text{ hours}$, constraining the origin release window to 2023-11-17T23:16:46+00:00 – 2023-11-17T23:24:16+00:00.

2. REVERSE-LAGRANGIAN HYDRODYNAMIC SIMULATION (OPENDRIFT UPSTREAM)

Reverse-Lagrangian simulation tracking 500 numerical particles through negative time steps driven by Copernicus Marine Service (CMEMS) reanalysis:

Step	Timestamp (UTC)	Hours Prior	Active Particles	Centroid Lat	Centroid Lon
Step 0	2023-11-17T23:54:16Z	T – 0h	500	28.92563°N	-88.96662°W
Step 1	2023-11-17T22:54:16Z	T – 1h	500	28.93537°N	-88.96780°W
Step 2	2023-11-17T21:54:16Z	T – 2h	500	28.94191°N	-88.96790°W
Step 3	2023-11-17T20:54:16Z	T – 3h	500	28.93486°N	-88.96207°W
Step 4	2023-11-17T19:54:16Z	T – 4h	500	28.93307°N	-88.96370°W
Step 5	2023-11-17T18:54:16Z	T – 5h	500	28.93311°N	-88.96757°W
Step 6	2023-11-17T17:54:16Z	T – 6h	500	28.93156°N	-88.96285°W

3. SUBSEA PIPELINE & OFFSHORE INFRASTRUCTURE ASSESSMENT

Infrastructure Parameter	Observed Cadastre Record	Forensic Context & Significance
Nearest Infrastructure	Subsea Pipeline	Bureau of Ocean Energy Management (BOEM) Record
Distance to Origin	48.7 meters	Co-located within physical dispersion boundary
Segment / Asset ID	Segment 12712	Operating Entity: WALTER OIL & GAS CORPORATION
Pipeline Specification	Diameter: 06" Status: ABN	Product Stream: BLKG (Bulk Gas / Hydrocarbon)

4. MARITIME TRAFFIC KINEMATICS & CANDIDATE VESSEL EVALUATION (AIS DOWNSTREAM)

All vessels within the 4D spatio-temporal query corridor evaluated via 1-minute resampled great-circle kinematics:

#	Vessel Name	MMSI	Vessel Type	CPA Dist	CPA Time	SOG	Score
1	CG WALNUT	366953000	Offshore / Tug / Other (90)	8451 m	23:18:55	13.3 kt	52.5%
2	KARLA F	367332840	Offshore / Tug / Other (90)	10914 m	23:44:30	9.8 kt	51.7%
3	MATTERHORN TLP	367376530	Offshore / Tug / Other (90)	25473 m	23:30:19	0.1 kt	50.3%
4	MR SEAMAN	367154130	Passenger (60)	10723 m	23:40:24	1.9 kt	47.5%
5	FULL CIRCLE II	338444731	Fishing / Trawler (37)	31965 m	22:59:59	15.2 kt	43.7%
6	GO GLORY	368269780	Cargo (70)	14243 m	23:02:25	0.1 kt	43.1%
7	NGUYEN T J	367594140	Fishing / Trawler (30)	14976 m	23:58:45	7.6 kt	42.7%
8	MISSISSIPPI III	367724890	Fishing / Trawler (30)	16704 m	23:17:50	7.6 kt	41.0%
9	BRETON ISLAND	367558730	Offshore / Tug / Other (90)	13754 m	22:17:14	2.5 kt	39.9%
10	BOOGIE	338176375	Fishing / Trawler (37)	33202 m	22:58:06	28.5 kt	39.8%
11	KIMBERLY CELESTE	368152930	Fishing / Trawler (30)	14212 m	22:17:19	6.0 kt	38.6%
12	JAMAL	366701330	Fishing / Trawler (37)	33731 m	22:59:38	0.1 kt	37.6%
13	A.B. CENAC	368238290	Unspecified / Auxiliary (0)	24372 m	22:17:41	1.4 kt	36.1%
14	OUR MOTHER	367155810	Fishing / Trawler (30)	28566 m	23:32:49	2.8 kt	35.6%
15	ENTERPRISE	368132210	Fishing / Trawler (37)	33729 m	22:17:54	0.0 kt	31.1%

5. CANDIDATE FORENSIC AUDIT PROFILES & NEGATIVE CONTROL EXONERATION

Commercial Surface Traffic Exoneration Summary

Exoneration Justification: All detected commercial vessels maintained a minimum Closest Point of Approach clearance of **>8.4 km to 33.7 km** from the physical release origin. Their low composite suspicion scores (31.1% – 52.5%) combined with continuous transponder broadcast feeds (0 blackout gaps) **conclusively clears all passing surface vessels of responsibility.**

Because the physical origin coincides within 309.4 meters of the known Taylor Energy MC-20 wellhead structure, the observed discharge is verified as an **ongoing infrastructure failure / subsea wellhead emanation** rather than an illicit surface vessel bilge dump.

6. MULTI-CRITERIA DECISION ANALYSIS (MCDA) SCORING FORMULATION

Evaluation Criterion	Weight (\$W\$)	Mathematical Decay Function & Forensic Basis
Spatial Proximity (S_{prox})	30%	Gaussian decay: $\exp(-(d_{cpa} / 2000)^2)$ — Rapid spatial penalization beyond 2.0 km
Temporal Alignment (S_{time})	30%	Continuous Gaussian decay centered on Fickian release window apex
Vessel Hazard Profile (S_{type})	15%	Regulatory multipliers: Tankers 1.0, Cargo 0.65, Tug/OSV 0.40, Passenger 0.20, Fishing 0.15
Slick Axis Alignment (S_{align})	15%	Angular cosine similarity: $\cos(\theta_{vessel} - \theta_{slick})$
Dark-Ship Gap Penalty (S_{gap})	10%	Step escalation for unannounced AIS transponder blackouts exceeding 30 minutes